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ICE 9B.1



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1. (2 marks) Consider the MAC algorithm $\text{MAC}(K, M)$ based on Triple-DES that uses a pair of two 56-bit keys, or (k_1, k_2) . The resulting MAC value so generated is 64 bits. H is some cryptographically strong hash function that produces a 64-bit hash value given a message of any length. Will this work?

- Yes (correct)
- No
- Cannot Say

2. (3 marks) Return to question 1 above. What is the time complexity of brute force attack to discover the key K in the above MAC?

- $O(2^{56})$
- $O(2^{64})$
- $O(2^{112})$ (correct)
- $O(2^{128})$

3. (3 marks) How many pairs of data must you possess before the above brute force attack can be launched?

- 1
- at most 3 (correct)
- at most 2^4
- at least 2^{64}
- at least 2^{128}
- None of the above

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Your work

Graded



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